

INMO(16th) – 2001

Time – 3 Hrs

Instructions:

- Calculators (in any form) and protractors are not allowed.
 - Ruler and compass allowed.
 - Answer all the questions.
1. Let ABC be a triangle in which no angle is 90° . For any point P in the plane of the triangle, let A_1, B_1, C_1 denotes the reflection of P in the sides BC, CA and AB respectively. Prove the following statements:
 - (a) If P is the incentre or an excentre of ABC , then P is the circumcentre of $A_1B_1C_1$.
 - (b) If P is the circumcentre of ABC , then P is the orthocentre of $A_1B_1C_1$.
 - (c) If P is the orthocentre or an excentre of ABC , then P is either the incentre or an excentre of $A_1B_1C_1$.
 2. Show that the equation $x^2 + y^2 + z^2 = (x - y)(y - z)(z - x)$ has infinitely many solutions in integers x, y and z .
 3. If a, b, c are positive real numbers such that $abc = 1$, prove that $a^{b+c}b^{c+a}c^{a+b} \leq 1$.
 4. Given any nine integers show that it is possible to choose, from among them, four integers a, b, c, d such that $a + b - c - d$ is divisible by 20. Further show that such a selection is not possible if we start with eight integers instead of nine.
 5. Let ABC be a triangle and D be the midpoint of side BC . Suppose $\angle DAB = \angle BCA$ and $\angle DAC = 15^\circ$. Show that $\angle ADC$ is obtuse. Further if O is the circumcentre of ADC , prove that triangle AOD is equilateral.
 6. Find all functions $f : \mathfrak{R} \rightarrow \mathfrak{R}$ such that $f(x + y) = f(x)f(y)f(xy)$ for all $x, y \in \mathfrak{R}$, where $\mathfrak{R}$ denotes the set of all real numbers.

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